Comparative Evaluation of Mandate-Constrained TD3 Models for Dynamic Portfolio Allocation

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May 29, 2026

Abstract

This paper evaluates whether Twin Delayed Deep Deterministic Policy Gradient (TD3) can produce mandate-credible dynamic portfolio allocations when the investment problem is judged under mandate constraints rather than by return statistics alone. The study compares eight state-specification families and their constrained variants using expanding walk-forward out-of-sample evaluation, multiple random seeds, transaction costs, turnover, drawdown, concentration diagnostics, static and dynamic benchmarks, and a conservative bootstrap validation layer. The main empirical question is whether TD3-based dynamic allocation can become mandate-credible under realistic investment constraints. A central diagnostic test is whether max-weight caps correct the concentration failure mode observed in unconstrained policies.

The evidence supports the latter interpretation. Unconstrained TD3 policies tend to collapse into concentrated allocations, while max-weight constraints materially improve diversification, drawdown behavior, and mandate-aware performance. The leading candidate, `V3_real_macro_vintage_clean_no_dxy_cap_0.50`, uses clean real-time/as-of FRED macro information consisting of DGS10, DGS2, VIX, and CPI. The preferred macro state deliberately excludes the dollar proxy because no full-window fresh true-vintage dollar series is available for 2015–2026 without fallback, discontinuation, or current-vintage relabeling. The GARCH-augmented extension, `V7_real_macro_vintage_clean_no_dxy_garch_cap_0.50`, remains competitive but does not outperform the simpler clean macro specification. Profile-specific reporting further shows that the preferred cap is mandate-dependent.

The contribution is therefore not an architectural novelty claim and not a claim that TD3 beats the market. It is a comparative, mandate-aware evaluation showing that action-space concentration constraints can stabilize deep reinforcement learning allocation behavior and make the strongest specifications competitive under client-mandate criteria. Bootstrap validation does not establish statistically clear superiority over clean benchmark rules, so the defensible result is mandate-aware stabilization rather than statistical dominance.

1 Introduction

Dynamic portfolio allocation is a natural application for continuous-control reinforcement learning, but it also exposes a practical weakness of unconstrained deep reinforcement learning: policies can learn concentrated allocations that look attractive in isolated backtests while violating realistic investment mandates. Much of the DRL portfolio literature emphasizes return, volatility, or reward-based performance. Less often are concentration degeneracy, mandate compliance, clean real-time macro timing, multi-seed walk-forward robustness, and cautious benchmark validation audited together. This paper studies that combined evaluation problem for TD3-based dynamic allocation.

The central empirical claim is deliberately narrow. The evidence does not show that TD3 universally dominates simple or dynamic benchmarks. Instead, the evidence shows that unconstrained TD3 tends to produce fragile, concentrated policies, while max-weight constraints materially improve diversification, drawdown behavior, and mandate-aware performance. Importantly, the paper's central question is not whether TD3 maximizes standalone backtest performance, but whether it can produce allocation policies that remain credible once concentration, drawdown, turnover, and benchmark constraints are imposed. As a diagnostic step, the analysis tests whether single-asset concentration reflects useful selection or a concentration failure mode. The cap experiments indicate that it was not: once realistic concentration limits are imposed, the risk-benefit tradeoff improves rather than deteriorates. The leading model is `V3 clean no-DXY cap 0.50`, a constrained TD3 policy using real-time/as-of FRED macro information for Treasury yields, equity volatility, and inflation. In the experiment outputs, this corresponds to `V3_real_macro_vintage_clean_no_dxy_cap_0.50`.

This framing matters because portfolio learning is not only a prediction problem. A policy that earns high returns by implicitly becoming a single-asset bet is not equivalent to a mandate-compliant allocation process. The evaluation therefore combines conventional performance statistics with turnover, transaction costs, maximum drawdown, effective number of assets, average maximum weight, robust score, and a mandate-aware score.

Research question. Can TD3-based dynamic portfolio allocation become mandate-credible under realistic concentration, drawdown, turnover, and benchmark constraints once alternative financial, macroeconomic, volatility, and regime-state specifications are evaluated under a common walk-forward, multi-seed protocol?

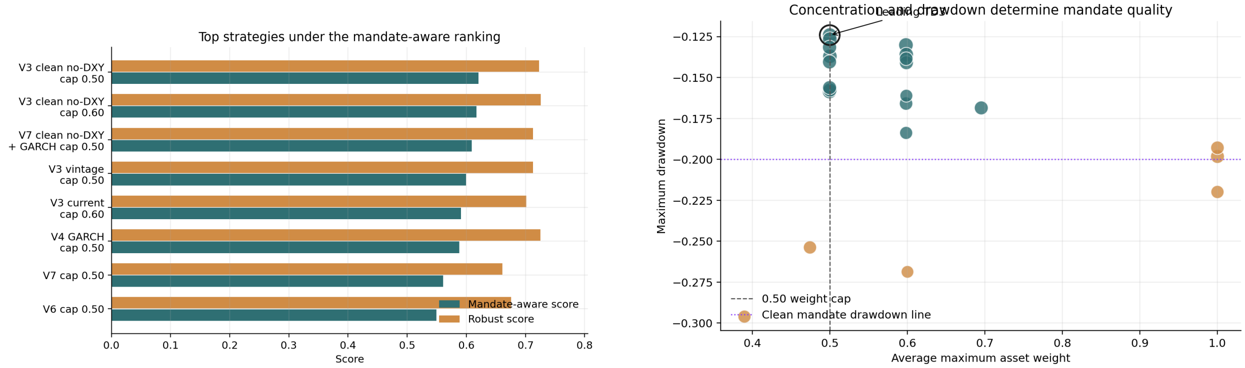
Contributions. First, the paper provides a comparative evaluation protocol for eight TD3 state-specification families, holding fixed the walk-forward timing, transaction-cost convention, benchmark set, concentration diagnostics, and multi-seed aggregation. Second, it documents a mandate-relevant failure mode: unconstrained TD3 can learn extreme allocations, and the cap experiments indicate that this concentration is not a robust source of policy quality. Third, it evaluates financial, macro, GARCH, macro-GARCH, regime, and EWMA/GARCH state designs without treating added feature complexity as evidence of improvement by itself. Fourth, it identifies a clean real-time/as-of macro state based on DGS10, DGS2, VIX, and CPI as the leading mandate-aware specification, explicitly excluding DXY because a consistent fresh true-vintage dollar proxy is unavailable for the full 2015–2026 window. Finally, it reports benchmark validation conservatively: the leading constrained TD3 variant ranks first by mandate-aware score, but bootstrap validation does not establish statistically clear superiority over clean benchmark rules.

Table 1: Positioning relative to closely related portfolio ML and DRL studies. The comparison emphasizes evaluation discipline rather than architectural novelty.

Paper	Method and data	Contribution	Position of this paper
A Deep Reinforcement Learning Framework for the Financial Portfolio Management Problem [Jiang et al., 2017]	Model-free DRL portfolio management in cryptocurrency markets	Direct portfolio-weight learning with transaction costs	Weekly multi-asset allocation with real-time macro states, mandate caps, and conservative benchmark validation
An Adaptive Portfolio Trading System [Almahdi and Yang, 2017]	Recurrent reinforcement learning with expected maximum drawdown in ETF trading	Drawdown enters the objective directly	TD3 allocation with hard max-weight constraints, effective-asset diagnostics, and macro/as-of information
A Novel Two-Stage Method for Well-Diversified Portfolio Construction Based on Stock Return Prediction Using Machine Learning [Chen et al., 2022]	ML return prediction plus diversified stock selection	Forecast-then-optimize pipeline with diversification reporting	Direct DRL allocation evaluated for concentration failure under client mandate limits
FinRL: A Deep Reinforcement Learning Library for Automated Stock Trading in Quantitative Finance [Liu et al., 2020]	Open-source DRL trading framework	Reproducible trading infrastructure	Specific constrained-TD3 research design rather than a library contribution
Asset Correlation Based Deep Reinforcement Learning for Portfolio Selection [Zhao et al., 2023]	Correlation-aware DRL with self-attention	Nonlinear cross-asset dependence modeling	Investor mandate constraints and clean real-time macro timing rather than only price-dependence modeling
Deep Reinforcement Learning for Portfolio Selection [Jiang et al., 2024]	TD3 with risk aversion and transaction costs on equity constituents	Closest TD3 benchmark	Explicit max-weight client constraints, no-DXY real-time macro discipline, concentration diagnostics, and cautious bootstrap interpretation
Deep Reinforcement Learning for High-Dimensional Multi-Period Portfolio Allocation [Jiang et al., 2025]	High-dimensional multi-period DRL allocation	Long-horizon allocation with costs and constraints	Smaller diversified universe where the research issue is mandate-aware credibility rather than scale

2 Related Literature

The empirical design connects three literatures. The first is modern portfolio selection, beginning with mean-variance optimization [Markowitz, 1952], dynamic portfolio choice [Merton, 1969, 1971], and the measurement of risk-adjusted performance [Sharpe, 1966]. The second is reinforcement learning and deep continuous control [Sutton and Barto, 2018], including deterministic policy gradients [Silver et al., 2014], DDPG [Lillicrap et al., 2015], and TD3 [Fujimoto et al., 2018]. The third is financial reinforcement learning and portfolio management, where prior work has applied recurrent reinforcement learning and deep agents to trading and allocation problems [Moody and Saffell, 2001, Almahdi and Yang, 2017, Jiang et al., 2017, Aboussalah et al., 2022, Jiang et al., 2024]. GARCH-style volatility modeling [Bollerslev, 1986] is used as an evaluated extension, while the statistical validation layer combines pairwise bootstrap comparisons with a White Reality Check style correction over the evaluated TD3 candidate set, following the broader concern that financial



(a) Top mandate-aware strategies

(b) Concentration and drawdown map

Figure 1: Overview: the leading constrained TD3 candidate ranks well while avoiding degenerate concentration.

model selection must account for data-snooping and backtest selection risk [White, 2000, Bailey and López de Prado, 2014]. López de Prado's *Machine Learning for Asset Managers* also motivates the methodological stance of this paper: machine learning should be used to test economically meaningful hypotheses and control backtest overfitting, rather than to search mechanically for high-performing trading rules [López de Prado, 2020].

This paper differs from performance-oriented backtests by treating mandate compliance as part of model quality rather than an afterthought. A high Sharpe ratio is not sufficient if the policy is effectively an unstable concentrated bet. The paper therefore evaluates constrained policies against transparent static and dynamic benchmarks under common timing, cost, and turnover conventions.

Relative to Jiang et al. [2024], the closest TD3 reference, this paper does not claim a new actor-critic architecture. Jiang et al. embed risk aversion and transaction-cost sensitivity in a TD3 portfolio-selection framework and evaluate large equity universes. Jiang et al. [2025] extend the same research program toward high-dimensional multi-period allocation. The present paper addresses a complementary question: in a smaller diversified asset universe, does TD3 remain mandate-credible once concentration limits, effective diversification, clean real-time macro timing, multi-seed walk-forward aggregation, and conservative benchmark validation are all imposed? The distinction matters because a reward penalty can reduce risk without ensuring that the realized portfolio history satisfies a client's concentration, turnover, and drawdown mandate.

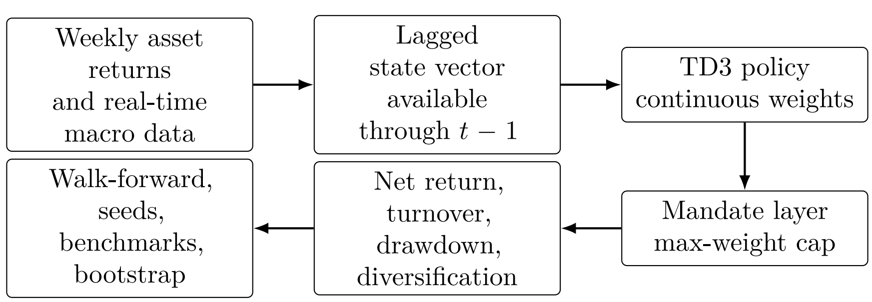


Figure 2: Empirical workflow and timing discipline. State variables available through $t-1$ determine the portfolio weights for week t; realized returns are observed only after the allocation is set.

3 Data and State Construction

The investable universe contains SPY, TLT, GLD, BTC-USD, and synthetic CASH. All standard experiments use weekly returns. CASH is treated as a zero-return asset in the standard protocol, so it functions as a defensive allocation channel rather than an estimated return process. The timing convention is lagged throughout: state information observed through week $t-1$ is used to choose weights for week t, and week-t returns are applied only after those weights have been fixed.

The leading macro specification uses real-time/as-of FRED vintage information for DGS10, DGS2, VIX, and CPI. Macro values are aligned to the weekly decision calendar using only information available by the allocation date. DXY is excluded from the preferred specification because no full-window fresh true-vintage dollar proxy was available for 2015–2026 without fallback, discontinuation, or current-vintage relabeling. This exclusion removes a potentially useful state variable, but it makes the preferred macro evidence cleaner by avoiding mixed-vintage labeling.

3.1 Investor Profile and Mandate Translation

The empirical design treats the client not as a return-maximizing abstraction but as an investor operating under a mandate. A mandate is a set of quantitative limits that translates risk tolerance into investable constraints: maximum drawdown, annualized volatility, single-asset concentration, diversification, and turnover. This is distinct from merely adding a risk-aversion coefficient to the reward. A risk-aversion coefficient changes the agent's objective; a mandate also defines which policies remain acceptable after training.

Table 2 reports the three mandate profiles implemented in the research infrastructure. The main reported candidate uses the stricter 0.50 concentration cap as the clean mandate anchor: drawdowns must remain economically acceptable, single-asset concentration must be controlled, and the policy should not become a hidden single-asset bet. Profile-specific reporting later shows that moderate and aggressive profiles may prefer the 0.60 variant among TD3 candidates. The reported mandate-aware score is therefore a research evaluation layer rather than a personalized investment suitability engine.

Table 2: Investor mandate profiles used to interpret policy quality. These profiles define evaluation constraints; they are not personalized investment recommendations.

Profile	Max drawdown	Max volatility	Max weight	Min eff. assets	Max turnover
Conservative	-0.10	0.15	0.50	2.00	0.50
Moderate	-0.20	0.25	0.80	1.25	0.75
Aggressive	-0.35	0.45	1.00	1.00	1.50

Source: mandate profile configuration in `src/risk/mandate_profiles.py`. The leading cap-0.50 result is especially relevant for concentration-aware clients because it enforces the conservative single-asset weight limit while remaining competitive under the final mandate-aware ranking.

Table 3: Data, assets, and feature families used in the reporting layer

Block	Variables	Role in the experiment
Investable universe	SPY, TLT, GLD, BTC-USD, CASH	Weekly allocation targets; CASH is synthetic zero-return in the standard protocol
Clean macro state	DGS10, DGS2, VIX, CPI	Real-time/as-of macro information used by the leading V3 clean no-DXY specification
Excluded macro proxy	DXY	Removed from the preferred specification because full-window fresh true-vintage coverage is not available without fallback, discontinuation, or relabeling
Risk and behavior diagnostics	Turnover, max drawdown, average max weight, effective assets	Used to distinguish investable mandate-aware behavior from concentrated backtest performance
Validation layer	Walk-forward folds, multiple seeds, static and dynamic benchmarks	Designed to reduce dependence on one split, one seed, or one benchmark family

3.2 Construction of the Eight Candidate State Specifications

The paper evaluates eight model families rather than a single preferred specification. The naming convention V1–V8 refers to the state representation given to the same TD3 allocation framework, with cap experiments applied after the unconstrained concentration behavior was diagnosed. This keeps the comparison focused: the research question is not whether one selected macro model works, but which information design survives a common mandate-aware evaluation protocol. Historical V3 variants that used current-vintage macro data or DXY fallback remain part of the audit trail, but they are not treated as final evidence once the clean no-DXY specification is available.

Table 4: Candidate state specifications evaluated before final model selection. V7 denotes the clean no-DXY V3 macro state augmented with GARCH features.

Family	State design	Construction logic
V1	Baseline financial	Lagged return, momentum, and volatility diagnostics from weekly asset returns
V2	Rich financial	Rolling momentum, EWMA volatility, beta/correlation versus SPY, drawdown variables, and market-regime flags
V3	Financial plus macro	V2 plus macro variables aligned by as-of timing; the final clean version uses DGS10, DGS2, VIX, and CPI
V4	Financial plus GARCH	V2 plus fitted GARCH-style volatility forecasts and related volatility diagnostics
V5	Regime ablation	Return-derived regime, correlation, and risk-off variables used to test simpler non-macro state blocks
V6	Parsimonious financial	Compact momentum/trend, defensive-asset attractiveness, volatility proxies, and risk-regime indicators
V7	Clean macro plus GARCH	V3 clean no-DXY macro state augmented with rolling fitted GARCH features
V8	EWMA/GARCH hybrid	V2 plus fitted GARCH, EWMA volatility, and comparison features between the two volatility estimates

The GARCH variants are constructed as rolling volatility forecasts rather than full-sample ex post features. For each eligible risky asset i, the fitted GARCH(1,1) forecast can be written as

$$\sigma^2_{i,t+1|t} = \omega_i + \alpha_i \epsilon^2_{i,t} + \beta_i \sigma^2_{i,t}, \tag{1}$$

where $\sigma^2_{i,t+1|t}$ is the one-step-ahead conditional variance, $\epsilon_{i,t}$ is the lagged innovation, and $(\omega_i, \alpha_i, \beta_i)$ are estimated using only information available before the allocation date. The implementation uses a rolling estimation window, requires a minimum return history before fitting, excludes synthetic CASH from fitted GARCH estimation, and falls back to realized rolling volatility only when the history is insufficient for a stable fit. The purpose of V4, V7, and V8 is therefore explicit: they test whether econometric volatility forecasts add value beyond return-derived and macro state variables.

4 Methodology

4.1 Portfolio Environment

At each weekly decision date, the agent observes a lagged state vector and selects portfolio weights over the asset universe. Weights are long-only and sum to one. The realized net financial return for strategy s in week t is

$$r^{\text{net}}_{s,t} = w^\top_{s,t} r_t - c \sum_{i=1}^{N} |w_{s,t,i} - w_{s,t-1,i}|, \tag{2}$$

where $w_{s,t}$ is the weight vector selected before week-t returns are realized, r_t is the realized asset-return vector, N is the number of assets, and c is the proportional transaction-cost rate. The turnover term uses the absolute change in asset weights between consecutive decision dates.

4.2 TD3 Agent

The agent is based on TD3, an actor-critic algorithm designed for continuous action spaces. TD3 addresses overestimation and instability in deterministic policy gradient methods through clipped double Q-learning, delayed policy updates, and target policy smoothing [Fujimoto et al., 2018]. In this application, the action is the portfolio weight vector, and the policy is evaluated by out-of-sample financial behavior rather than by training reward alone.

There is not one universal learned policy in the experiment. A separate policy is trained for each candidate state specification, walk-forward fold, random seed, and cap setting. Reported tables aggregate the resulting out-of-sample histories so that the selected strategy represents a model family plus mandate constraint, not a single lucky training run.

4.3 Mandate Constraints

The key intervention is a maximum asset-weight cap. For the leading clean macro model, the selected cap is 0.50. This constraint is not introduced as a post-hoc cosmetic adjustment; it represents a realistic mandate limit designed to prevent the agent from satisfying the objective through single-asset concentration. The evaluation also reports average maximum weight and effective number of assets to verify whether the constraint changes behavior in practice.

This treatment is intentionally more operational than a generic utility function. In a client-facing setting, an allocation with attractive realized return can still be rejected if it violates concentration or drawdown limits. The cap therefore acts as a hard feasibility layer, while the mandate-aware score acts as a reporting layer that ranks only economically acceptable behavior. The design does not claim to solve all aspects of suitability; it formalizes the part of the mandate that can be audited from portfolio histories.

4.4 Robust and Mandate-Aware Scores

The robust score is a normalized composite metric designed to avoid selecting models from one attractive statistic alone. It is an evaluation and reporting metric, not the TD3 training objective unless explicitly stated in a given experiment. Let each component be scaled to the unit interval, where larger values are better. For strategy s,

$$\begin{aligned}\mathrm{RobustScore}_s = {} & 0.30\,\mathrm{DSR}_s + 0.20\,\mathrm{SortinoScore}_s + 0.20\,\mathrm{CalmarScore}_s \\ & + 0.15\,\mathrm{DrawdownScore}_s + 0.10\,\mathrm{StabilityScore}_s + 0.05\,\mathrm{DisciplineScore}_s. \end{aligned} \tag{3}$$

The discipline component summarizes investability diagnostics such as cash usage, turnover, and diversification. The metric is reported alongside raw Sharpe ratio, drawdown, and effective-assets values so that the composite score does not hide its economic drivers. Because DSR enters the robust score as one component, raw Sharpe statistics are not interpreted as standalone evidence of model superiority.

The mandate-aware score is also a research evaluation layer, not a personalized suitability engine. It applies a client-style drawdown feasibility filter to the robust score. Strategies are grouped as clean-mandate candidates when maximum drawdown is no worse than -20%, yellow candidates when drawdown is no worse than -25%, red candidates when drawdown is no worse than -30%, and ineligible otherwise. Let $\mathrm{MDD}_s \leq 0$ denote maximum drawdown for strategy s. The drawdown recovery burden is

$$R_s = \frac{|\mathrm{MDD}_s|}{1 - |\mathrm{MDD}_s|}, \tag{4}$$

and the drawdown multiplier is

$$m_s = \max(0, 1 - R_s). \tag{5}$$

Eligible strategies receive

$$\mathrm{MandateAwareScore}_s = \mathrm{RobustScore}_s \times m_s, \tag{6}$$

while ineligible strategies receive a score of zero. This is why a strategy with a high standalone Sharpe ratio can rank below a more diversified and less fragile candidate.

4.5 Experimental Protocol

The official comparison protocol uses expanding walk-forward validation with aligned returns, asset ordering, evaluation dates, transaction-cost rules, and turnover conventions. Static benchmarks include single-asset buy-and-hold strategies, equal-weight portfolios, and a 60/40 SPY-TLT portfolio. Dynamic benchmarks include lagged momentum, risk-adjusted momentum, trend-following SPY/CASH, defensive risk-off, rolling inverse-volatility risk parity, and rolling long-only Markowitz-style rules where available.

Table 5: Model families and reporting role. Final comparisons use clean no-DXY evidence where available; current-vintage and DXY-fallback variants are retained only as historical diagnostics.

Family	Preferred cap	Reporting role	Interpretation in the paper
V3 clean no-DXY macro	0.50	Final candidate	Leading mandate-aware candidate; clean real-time/as-of macro state without dollar fallback
V7 clean no-DXY + GARCH	0.50	Final extension	Tests whether fitted GARCH features improve over the clean macro state
V4 real GARCH current	0.50	Final comparison	Strong volatility-state candidate; useful benchmark for feature complexity
V2/V5/V6/V8 variants	Candidate-specific	Final comparisons	Supporting feature-set and cap-sensitivity comparisons across financial and volatility states
V3 vintage macro with DXY fallback	0.50	Historical diagnostic	Improved over current-vintage macro but superseded by clean no-DXY design
V3 current-vintage macro	0.60	Historical diagnostic	Useful for tracing development, but not preferred evidence because it relies on current-vintage macro labeling

Table 6: Central mandate-aware ranking of selected final constrained candidates. The leading entry is the clean real-time/as-of no-DXY V3 specification with a 0.50 max-weight cap.

Rank	Strategy	Mandate	Robust	Sharpe	Max DD	Eff. assets	Role
1	`V3 clean no-DXY cap 0.50`	0.6207	0.7232	0.8393	-0.1241	3.1458	Leading clean macro
2	`V7 clean no-DXY + GARCH cap 0.50`	0.6093	0.7126	0.8319	-0.1266	3.1540	Macro + GARCH extension
3	`V4 GARCH cap 0.50`	0.5883	0.7250	0.7528	-0.1586	3.1347	Volatility-state candidate
4	`V6 financial state cap 0.50`	0.5492	0.6757	0.5941	-0.1576	3.1407	Parsimonious financial state
5	`V2 reference cap 0.50`	0.5482	0.6553	0.8253	-0.1405	3.1109	Reference feature set
6	`V8 EWMA+GARCH cap 0.50`	0.5374	0.6595	0.5778	-0.1562	3.1255	Volatility hybrid
7	`V5 no-vol block cap 0.70`	0.5294	0.6641	0.8568	-0.1686	1.9457	Ablation/regime comparison

Historical V3 current-vintage and DXY-fallback variants remain part of the research audit trail, but they are excluded from this final-candidate table because the clean no-DXY design supersedes them as evidence. Source: final constrained TD3 report outputs.

5 Results

The results are organized around the research hierarchy: first, whether constrained TD3 becomes mandate-credible; second, whether max-weight caps correct the concentration failure mode; and third, whether additional state complexity improves on the clean macro specification.

5.1 Main Mandate-Aware Ranking

Table 6 is the central empirical table. It reports the selected final constrained candidates by model family using the final clean reporting layer. The preferred model is `V3 clean no-DXY cap 0.50`, with a mandate-aware score of 0.6207, robust score of 0.7232, Sharpe ratio of 0.8393, maximum drawdown of -0.1241, and average effective number of assets of 3.1458. It ranks above the GARCH-augmented clean macro candidate, `V7 clean no-DXY + GARCH cap 0.50`, which is competitive but does not outperform the simpler clean macro state. The V4 GARCH candidate has a strong robust score of 0.7250, but its larger drawdown lowers its mandate-aware score relative to the clean V3 leader.

5.2 Concentration and Drawdown Behavior

The max-weight cap materially changes the economic character of the learned policy. The leading clean macro candidate has average maximum weight close to 0.50 by construction and an effective number of assets above 3, in contrast to uncapped candidates that tend toward near-single-asset concentration. The key result is therefore behavioral as well as statistical: constraints reduce a degenerate allocation pattern that would otherwise make the learned policy less credible for mandate-aware portfolio management.

Table 7: Cap-sensitivity summary for the clean macro candidates. This table is a cap-sensitivity and seed-replication robustness check, not the main final ranking source.

Candidate	Best cap	Mandate	Robust	Max DD	Interpretation
V3 clean no-DXY macro	0.50	0.6035	0.7050	-0.1259	Cap improves mandate behavior
V7 clean no-DXY + GARCH	0.50	0.6093	0.7126	-0.1266	Competitive, not superior to V3

Source: cap-sensitivity summaries for `cap_sensitivity_experiment_v3_clean_no_dxy_60ep_10seeds_replication` and `cap_sensitivity_experiment_v7_clean_no_dxy_garch_60ep_10seeds`. The V3 clean no-DXY values come from the replication seed grid. The V7 values come from the corresponding clean no-DXY GARCH cap-sensitivity run. The replication run confirms the same qualitative cap benefit with a different seed grid, but its V3 values should not be mixed mechanically with the final main-report values in Table 6.

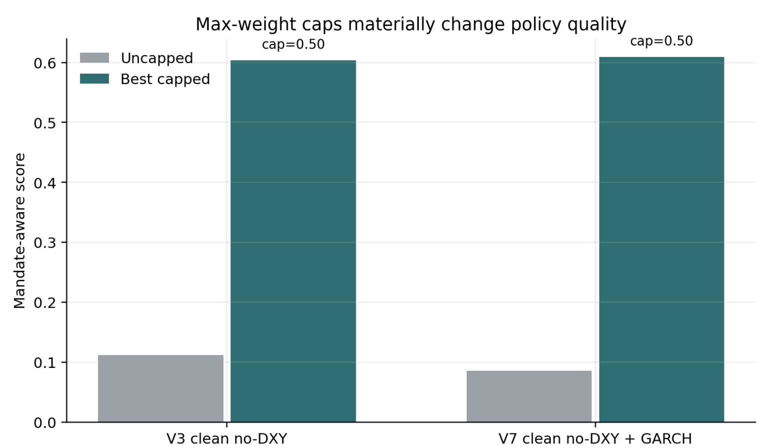
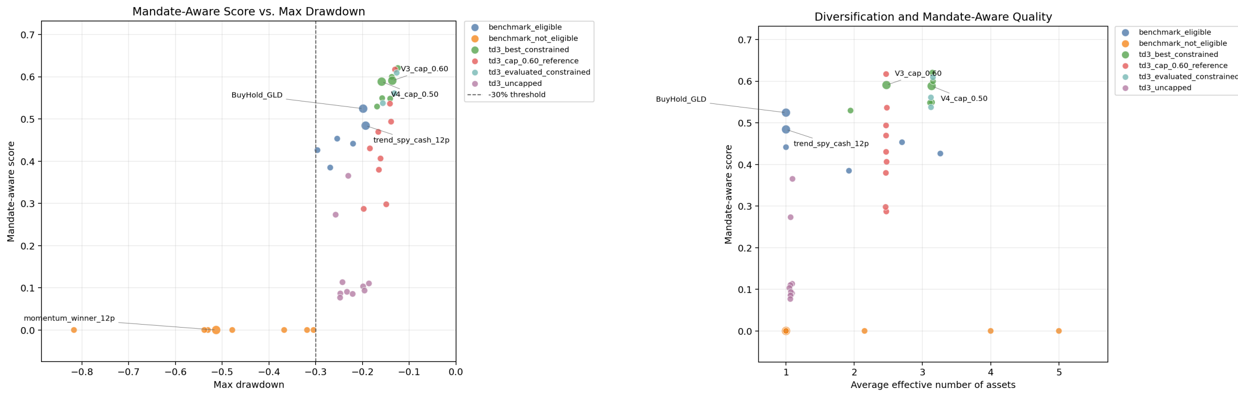


Figure 3: Max-weight caps improve the mandate-aware score profile.



(a) Mandate score and drawdown

(b) Diversification and mandate score

Figure 4: Mandate-aware performance versus drawdown and diversification.

5.3 Benchmark Comparisons

The leading constrained TD3 candidate beats the best clean benchmark (`BuyHold_GLD`) by mandate-aware score in the final report. However, the broader benchmark picture remains nuanced. Some aggressive benchmarks achieve higher robust scores but fail drawdown eligibility, while clean benchmarks remain difficult references in statistical validation. The evidence therefore supports competitiveness under mandate-aware evaluation, not unconditional benchmark dominance.

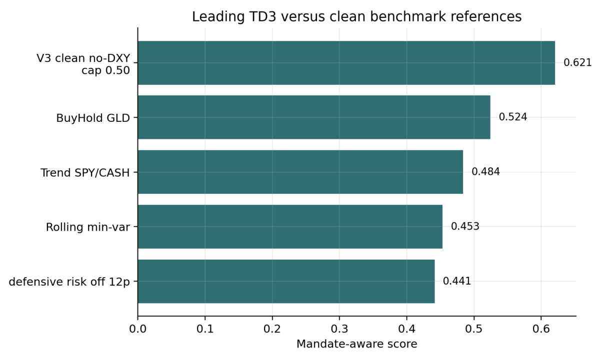
Table 8: Selected clean benchmark comparisons. The leading constrained TD3 candidate ranks above these clean benchmarks by mandate-aware score, but this table is not a statistical dominance test.

Strategy	Mandate	Robust	Sharpe	Max DD	Eff. assets	Role
V3 clean no-DXY cap 0.50	0.6207	0.7232	0.8393	-0.1241	3.1458	TD3 candidate
BuyHold GLD	0.5244	0.6967	0.8265	-0.1983	1.0000	Best clean benchmark
Trend SPY/CASH 12p	0.4841	0.6362	0.8065	-0.1929	1.0000	Dynamic benchmark
Rolling min-variance 52p	0.4533	0.6870	0.9229	-0.2538	2.7008	Diversified benchmark

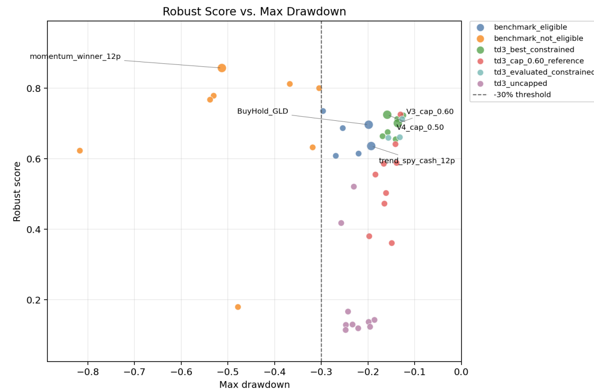
Roles are shortened for readability. The table compares mandate-aware ranking levels only; statistical validation is reported separately in Table 9.

5.4 Statistical Validation

Table 9 summarizes the conservative bootstrap comparison against clean benchmark strategies. The leading constrained TD3 model is not statistically clear versus `BuyHold_GLD` or the trend-following SPY/CASH benchmark. Against `BuyHold_GLD`, the Sharpe delta is -0.7796 with a 95% interval from -1.7984 to 0.2417, and the bootstrap probability that the candidate beats the benchmark is 0.0940. Against `trend_spy_cash_12p`, the delta is -0.1202 with a 95% interval from -0.9890 to 0.7314, and the bootstrap probability that the candidate beats the benchmark is 0.4110. These results do not prove TD3 superiority over clean benchmarks. They do not invalidate the mandate-aware



(a) Mandate-aware benchmark gap



(b) Robust score and drawdown

Figure 5: Benchmark gap and robust-score/drawdown tradeoff.

result, because the paper's claim is behavioral stabilization and mandate-aware competitiveness, not statistical dominance.

Table 9: Bootstrap validation against clean benchmarks. The probability column reports the bootstrap probability that the candidate's Sharpe ratio exceeds the benchmark's Sharpe ratio; it is not reported as a classical p-value.

Candidate	Benchmark	Sharpe delta	95% CI lower	95% CI upper	Prob. candidate beats
`V3 clean no-DXY cap 0.50`	`BuyHold_GLD`	-0.7796	-1.7984	0.2417	0.0940
`V3 clean no-DXY cap 0.50`	`trend_spy_cash_12p`	-0.1202	-0.9890	0.7314	0.4110
`V7 clean no-DXY + GARCH cap 0.50`	`BuyHold_GLD`	-0.7869	-1.8855	0.2610	0.1020
`V7 clean no-DXY + GARCH cap 0.50`	`trend_spy_cash_12p`	-0.1290	-0.9613	0.6898	0.4000

The bootstrap layer date-averages TD3 fold/seed histories before resampling so overlapping dates are not treated as independent observations. Confidence intervals are approximate and depend on the selected block length.

Table 10: White Reality Check against clean benchmarks

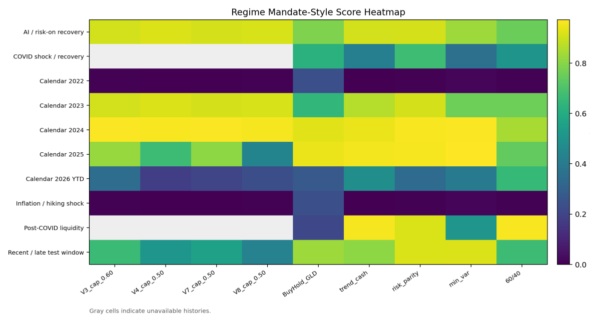
Benchmark	Candidates	Periods	Best searched TD3	Weekly mean diff	WRC p-value	Interpretation
`BuyHold_GLD`	10	228	V6 cap 0.50	-0.001891	0.9065	No evidence of searched TD3 superiority
Trend SPY/CASH 12p	10	228	V6 cap 0.50	0.000765	0.4503	No evidence of searched TD3 superiority

Source: `outputs/tables/white_reality_check_final/`. The White Reality Check is computed over aligned realized out-of-sample weekly net returns and controls for selecting the best candidate among the evaluated TD3 set. It tests mean return differentials, not mandate-aware score. High p-values indicate no evidence that the best searched TD3 candidate is superior to the benchmark after accounting for model search.

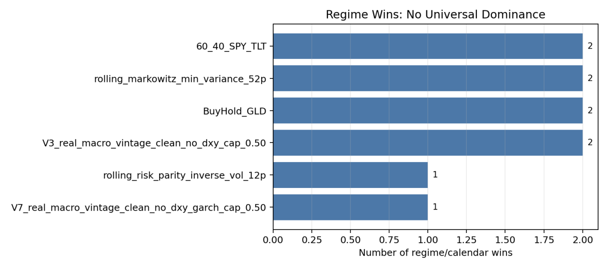
The White Reality Check strengthens the conservative interpretation. After accounting for the search across evaluated TD3 candidates, the best candidate by mean return differential does not show statistically reliable superiority over `BuyHold_GLD` or Trend SPY/CASH 12p. This does not invalidate the mandate-aware result because the paper's central claim is not mean-return dominance. The claim is that max-weight constraints reduce degenerate concentration and make TD3 policies more credible under mandate-aware evaluation.

5.5 Regime Sensitivity

The regime analysis reinforces the conservative interpretation. There is no universal winner across market environments. Constrained TD3 candidates become competitive in several out-of-sample mandate-aware windows, but benchmark rules dominate other slices. Some early regimes, including COVID and 2021, have limited TD3 test-history coverage under the current walk-forward output structure; this is a limitation of the available regime panel rather than evidence of regime-level dominance. The appropriate conclusion is regime-sensitive competitiveness, not universal superiority.



(a) Mandate score by regime



(b) Regime winners

Figure 6: Regime sensitivity and absence of a universal winner.

5.6 Mandate Profile Sensitivity

The mandate-profile comparison is a profile-specific evaluation over the already trained candidate grid and realized histories. It is not a mandate-conditioned training experiment; its purpose is to test whether the preferred policy changes when the same realized histories are scored under different mandate profiles. This reporting layer does not replace the main clean mandate result in Table 6.

Table 11: Mandate-profile sensitivity of the overall winner. The preferred strategy changes as mandate constraints become less restrictive.

Profile	Overall winner	Profile score	Robust	Max DD	Eff. assets	Avg max wt	Turnover	Interpretation
Conservative	V3 clean no-DXY cap 0.50	0.5738	0.7232	-0.1241	3.1458	0.4997	0.2010	Stricter concentration control remains preferred
Moderate	V3 clean no-DXY cap 0.60	0.7257	0.7257	-0.1302	2.4676	0.5980	0.2887	Cap 0.60 improves the profile score while remaining admissible
Aggressive	Equal Weight	0.8004	0.8004	-0.3044	5.0000	0.2000	0.0000	Benchmark behavior becomes admissible and wins overall

Source: `outputs/tables/mandate_profile_comparison_final/`. Full strategy identifiers are retained in the generated output files. Profile score applies the corresponding mandate thresholds to the already realized strategy metrics; higher values indicate better profile-specific admissibility and quality. V7 clean no-DXY + GARCH remains competitive in the broader report but does not displace V3 clean no-DXY in these profile-specific winners. The table is descriptive and does not imply statistical superiority.

The sensitivity analysis confirms that mandate credibility is profile-dependent. The conservative profile preserves V3 clean no-DXY cap 0.50 as the preferred clean mandate anchor. Moderate and aggressive TD3 rankings prefer the cap-0.60 V3 clean no-DXY variant, while the aggressive overall ranking favors Equal Weight. This does not change the main result: max-weight constraints stabilize TD3 allocation behavior, but the preferred cap depends on the mandate under which realized histories are evaluated.

6 Discussion

The findings answer the research hierarchy in a qualified way: TD3 becomes mandate-credible only after concentration constraints are imposed; the unconstrained concentration pattern behaves more like a failure mode than useful selection; and additional GARCH complexity does not outperform the simpler clean macro state. This interpretation is consistent with López de Prado's view that the value of machine learning in asset management lies less in black-box backtest maximization and more in disciplined feature design, theory discovery, and robustness against false investment strategies.

The results point to a practical lesson for financial reinforcement learning: action-space constraints can matter as much as state-space sophistication. More features do not automatically produce a better allocation policy. The GARCH-augmented clean macro model remains competitive, but it does not improve on the simpler clean V3 specification in the final mandate-aware ranking. The strongest evidence emerges when the allocation problem is constrained in a way that resembles a realistic client mandate.

The clean no-DXY macro design is also a methodological strength rather than only a loss of information. DXY is an economically plausible feature, but using it without a consistent full-window fresh true-vintage series would mix timing conventions or rely on fallback labeling. Removing it narrows the state representation while making the preferred empirical claim cleaner and easier to audit.

7 Limitations

Several limitations remain. First, bootstrap validation and the White Reality Check do not establish statistically clear superiority over clean benchmarks; the White Reality Check is applied to mean return differentials, not mandate-aware score. Second, the results are regime-sensitive, and some early regimes have limited TD3 histories under the available walk-forward output structure. Third, the mandate-aware score is a constructed research evaluation layer and should be reported alongside its components rather than interpreted as a universal welfare or suitability measure. Fourth, although the robust-score construction includes a DSR component where available, this manuscript does not yet report a standalone Deflated Sharpe table for every final candidate. A full DSR appendix across all searched strategies is left for the extended statistical appendix. Fifth, the asset universe is small and diversified; the results do not address high-dimensional single-equity allocation at scale. Sixth, the transaction-cost model is simplified. Seventh, mandate-profile selection is evaluated over the trained candidate grid; mandate-conditioned training, SPA-style testing, broader multiple-testing corrections, or nested validation-based profile selection are left for future work. Finally, the results are not a production trading strategy, investment recommendation, or deployable alpha claim.

8 Conclusion

This paper evaluates whether TD3-based dynamic allocation can become mandate-credible under realistic investment constraints after comparing eight financial, macro, volatility, and regime-state specifications. The answer is qualified. Unconstrained TD3 remains fragile because it tends toward excessive concentration. Max-weight constraints materially improve allocation behavior and make the leading clean macro TD3 candidate credible under the mandate-aware evaluation layer, while the GARCH extension remains competitive but not superior. Mandate credibility is profile-dependent: conservative reporting prefers the stricter cap-0.50 clean V3 candidate, moderate and aggressive TD3 profiles admit the cap-0.60 variant, and the aggressive overall ranking can favor a benchmark. Benchmark comparisons and bootstrap validation require caution: the evidence supports a defensible constrained-allocation finding, not a universal claim that TD3 beats the market. White Reality Check results reinforce the same caution: the searched TD3 candidate set does not statistically dominate clean benchmarks by mean return differential. The result is not market dominance, but mandate-aware stabilization of DRL allocation behavior.

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